

Time: 180 Minutes

Max. Marks: 50

- Instruction:**
1. In case of any doubt, write your assumptions, write it clearly and continue.
 2. There are 06 questions printed on the paper. Attempt any 5 questions.
 3. Please write all sub-parts of a question in a sequence.

- Q1. (a) Why loss function is chosen as squared loss instead of absolute or actual loss in regression problems? **02**
(b) Design a McCulloch Pitts neuron to model a 3-Bit logical AND problem. **02**
(c) Explain the weight updating equations for the following algorithms: **04**
i) Adagrad ii) RMSProp
(d) What are the characteristics of a sequential learning problem? **02**
- Q2. (a) Which of the following is/are true? If any statement is false, make it true without diverting from the context of the question. **04**
a. Simple models has high bias and high variance.
b. High bias and low variance are the desirable characteristic of a good model.
c. Variance contribute to error while Bias do not contribute to error.
d. In NAG optimizer, gradient term is first added to the old weight and then momentum term is added.
(b) Consider a convolutional neural network block whose input size is $64 \times 64 \times 8$. The block consists of the following layers: **06**
• A convolutional layer 32 filters with height and width 3 and 0 padding which has both a weight and a bias (i.e. CONV3-32)
• A 2×2 max-pooling layer with stride 2 and 0 padding (i.e. POOL-2)
• A batch normalization layer (i.e. BATCHNORM)
Compute the output activation volume dimensions and number of parameters of the layers. You can write the activation shapes in the format ($H; W; C$ where $H; W; C$ are the height, width, and channel dimensions, respectively.
i) What is the output activation volume dimensions and number of parameters for CONV3-32?
ii) What is the output activation volume dimensions and number of parameters for POOL2?
iii) What is the output activation volume dimensions and number of parameters for BATCHNORM?
- Q3. (a) Given a recurrent neural network for a sequential problem with k -dimensional inputs, n -dimensional states and d -classes. If U is the weight matrix between input to states, V is the weight matrix between states to outputs and W is the weight matrix between states then what is the dimension of U, V and W ? Mention the equations of state s_i and output y_i . **03**
(b) You are using an Adam optimizer. Show why the bias correction naturally disappears when the numbers of steps to compute the exponential moving averages gets large. **03**
(c) What are the Gates used in standard LSTM model? Explain the Gates mathematically. Why GRU is computationally cheaper than LSTM? **04**
- Q4. (a) Why an activation function is used in neural network? What are the characteristics of a good activation function? Explain the merits and drawbacks of sigmoid activation function by computing its derivative? **04**
(b) You decide to build an image classifier to predict whether a watermelon is sweet (1) or not (0). **06**
i) You've built your own labeled dataset, model and are thinking about using the mean squared error (MSE) loss to optimize model parameters. Give one reason why MSE might not be a good choice for your loss function. [1 Mark]
ii) You decide to use the binary cross-entropy (BCE) loss function. Write down the formula for this loss (for a single example) in terms of the label y and prediction y' . (1 formula) [1 Mark]
iii) What value does the loss take for a prediction of $y' = 0.9$ on a negative ($y = 0$) example. You can simply fill in the formula with the numbers, without needing to calculate it. (1 formula). [1 Mark]

iv) Compute the total cost, J , of the network averaged across the following dataset of 3 examples using the binary cross entropy loss. $Y^T = (1, 0, 0)$, and $Y^T = (0.1, 0.2, 0.7)$. (1 formula). [3 Marks]

Q5. (a) You come up with a CNN classifier. For each layer, calculate the number of parameters and the dimension of the associated feature maps (input). The notation follows the convention: 05

- CONV- K - N denotes a convolutional layer with N filters, each them of size $K \times K$, Padding and stride parameters are always 0 and 1 respectively.
- POOL- K indicates a $K \times K$ pooling layer with stride K and padding 0.
- FC- N stands for a fully-connected layer with N neurons.

Layer	Feature map dimension	Number of parameters
INPUT	$128 \times 128 \times 3$	0
CONV-9-32		
POOL-2		
CONV-5-64		
POOL-2		
CONV-5-64		
POOL-2		
FC-10		
FC-3		

(b) What is the number of perceptron units that are sufficient to model the XOR function with 3 variables with only a single hidden layer and an output layer? Draw the model. 02

(c) You want to predict the next character in an auto completion sequential problem. You are using a recurrent neural network model to solve it. However, your model don't give better accuracy. 03

- i) Will shuffling your dataset improve the accuracy? Justify your answer in a single line?
- ii) What loss function will you choose for this problem?
- iii) What can you tell about the dimension of input sequence instances?

Q6. (a) The convolution operation is applied on an input image (as in Fig.1) using a filter as in Fig.2 with stride 1. What will be the output image? If max pooling is applied on the convoluted image with a 2×2 filter and stride 1 what will be its resultant feature representation? If padding is to be used on the convoluted image to make its dimension same as input image what will be the padding size? 05

1	2	3	4	0	6
3	0	5	2	1	8
1	1	7	3	8	1
0	2	0	3	4	0
1	3	0	5	2	1
0	2	1	0	3	0

Fig. 1 Input Image

1	2	1
0	0	0
-1	-2	-1

Fig.2 Filter

(b) Explain guided Backpropagation algorithm to vizualize the inference of a neuron in convolutional neural network. 03

(c) Mention at least eight forms of regularization that can be used in deep learning. 02

*****End of Question Paper*****